

Array Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

Language-Agnostic DSA Problem Bank | Focus: Algorithmic thinking on sequences, not language-specific list methods.

Section 1: Array Internals and Basic Operations

- ★ Create an Array and Access Elements by Index
- ★ Traverse an Array (Forward and Backward)
- ★ Insert an Element at a Given Position
- ★ Delete an Element at a Given Position
- ★ Update an Element at a Given Index

Find Length of an Array without Built-in Function

Copy an Array without Built-in Methods

Merge Two Arrays into a Third Array

Section 2: Searching and Counting

- ★ Linear Search in an Array
- ★ Count Occurrences of an Element
- ★ Find Maximum and Minimum Element

Find Second Largest and Second Smallest

- ★ Find All Duplicate Elements

Find Missing Number in Array of 1 to N

Find the Element Appearing Once (Others Appear Twice)

Section 3: Sorting-Based Operations

- ★ Sort Array in Ascending Order (using any sorting logic)
- ★ Sort Array in Descending Order

Sort an Array of Strings by Length

- ★ Sort by Frequency of Elements

Section 4: Two Pointer Technique

- ★ Check if Array has a Pair with Given Sum (Sorted Array)
- ★ Find Pair with Given Sum (Unsorted Array)
- ★ Remove Duplicates from Sorted Array In-Place
- ★ Reverse an Array In-Place using Two Pointers

Three Sum Problem (Find Triplets with Given Sum)

Container with Most Water

Move All Zeros to End while Preserving Order

Section 5: Sliding Window Technique

- ★ **Maximum Sum Subarray of Size K**
- ★ **Minimum Sum Subarray of Size K**
- ★ **Longest Subarray with Sum Equal to K**

Longest Subarray with At Most K Distinct Elements

Count Occurrences of Anagrams in a String (Sliding Window)

Minimum Window Substring

Section 6: Prefix Sum and Difference Array

- ★ **Build Prefix Sum Array**
- ★ **Range Sum Query using Prefix Sum**
- ★ **Equilibrium Index of an Array**

Subarray with Zero Sum

Product of Array Except Self (without Division)

Range Update using Difference Array

Section 7: Subarray Problems

- ★ **Maximum Subarray Sum (Kadane's Algorithm)**
- ★ **Maximum Product Subarray**

Minimum Subarray Sum

- ★ **Longest Increasing Subarray**

Smallest Subarray with Sum Greater than X

Count Subarrays with Product Less than K

Section 8: Rotation and Rearrangement

- ★ **Left Rotate Array by K Positions**
- ★ **Right Rotate Array by K Positions**
- ★ **Rearrange Array: Positive and Negative Alternately**

Check if One Array is a Rotation of Another

Rearrange Array: Leaders (Elements Greater than All to Right)

Segregate Even and Odd Numbers

Section 9: Advanced Array Patterns

- ★ **Trapping Rain Water Problem**
- ★ **Jump Game (Can Reach Last Index?)**

Jump Game II (Minimum Jumps to Last Index)

- ★ **Find Peak Element**

Majority Element (Moore's Voting Algorithm)

Median of Two Sorted Arrays

Section 10: Real-World Applications of Arrays

- ★ **Stock Buy and Sell — Maximum Single Transaction Profit**
- ★ **Student Marks Analysis — Average, Highest, Lowest, Rank**

Stock Buy and Sell — Maximum Profit with Multiple Transactions

★ **Temperature Data Analysis — Moving Average using Sliding Window**

Sensor Reading Smoothing — Replace each Value with Average of Neighbours

★ **Scheduling Conflict Detection — Check if Time Slots Overlap**

Exam Seat Allocation — Arrange Students with Minimum Conflict Distance

Score Board Updater — Track Scores and Rank Players in Real Time